

EXPT NO :- 2 Verification of De-Morgans Law

THEORY:

According to De-Morgan Law :

$$(a) \overline{A + B} = \overline{A} \cdot \overline{B} \dots\dots\dots (2.1)$$

$$(b) \overline{A \cdot B} = \overline{A} + \overline{B} \dots\dots\dots (2.2)$$

Taking the expression (2.1) and separating it into two individual expressions we get

$$Y_1 = \overline{A + B} \dots\dots\dots (2.3)$$

$$Y_2 = \overline{A} \cdot \overline{B} \dots\dots\dots (2.4)$$

To construct a logic circuit to perform the function for the Boolean expression (2.3) we require a NOT gate and a OR gate. The logic circuit diagram for the expression (2.4) we require a AND gate and two NOT gates. The logic circuit diagram is shown in the figure (2.1)

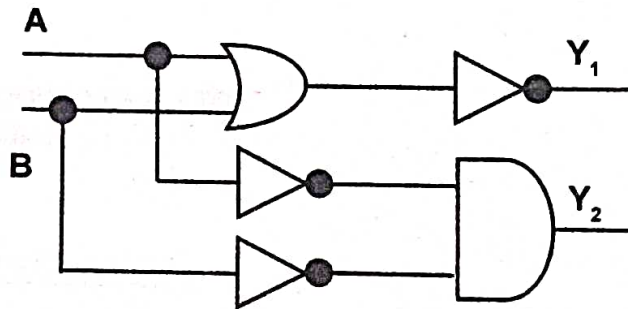


figure (2.1)

Now if Y_1 is equal to Y_2 are equal then the De-Morgan's first law is proved.

Taking the second expression and separating the expression (2.2) we get

$$Y_3 = \overline{A \cdot B} \dots\dots\dots (2.6)$$

$$Y_4 = \overline{A + B} \dots\dots\dots (2.7)$$

To construct the logic circuit for the expression (2.6), we require a AND gate and a NOT gate. To construct the logic circuit for the expression we require a OR gate and two NOT gates. The complete logic circuit diagram is shown in the figure(2.2)

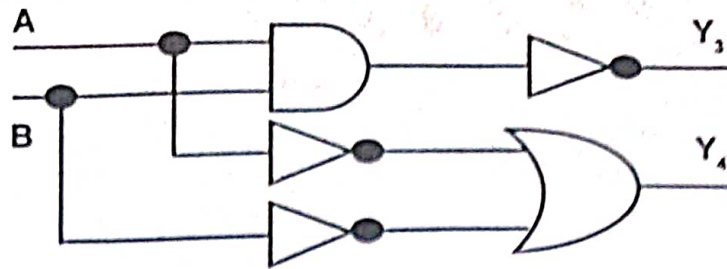


figure (2.2)

APPARATUS AND DEVICES REQUIRED:

- (a) Semiconductor IC
 - (i) Quad Two input AND gate : 74LS08
 - (ii) Quad Two input OR gate : 74LS32
 - (iii) Hex inverter : 74LS04
- (b) Jumpers
- (c) Bread Board
- (d) + 5 Volts Regulated Power supply.
- (e) Light emitting Diode (LED)

EXPERIMENTAL OBSERVATION:

(a) Truth Table for verification of De-Morgan first law:

Input logic A	Input logic Voltage	Input logic B	Input logic Voltage	Logic Output Y ₁	Logic Output Voltage	Logic Output Y ₂	Logic Output Voltage

Conclusion : We can see that the outputs are equal thus De-Morgans first law is proved.

(b) Truth Table for verification of De-Morgan first law: $\overline{A \cdot B} = \overline{A} + \overline{B}$

Input logic A	Input logic Voltage	Input logic B	Input logic Voltage	Logic Output Y ₃	Logic Output Voltage	Logic Output Y ₄	Logic Output Voltage

Conclusion : We can see that the outputs are equal thus De-Morgans Second law is proved.

DISCUSSION :

- (i) The bread makes circuit connections highly flexible. There is no need for soldering or binding screws.
- (ii) The light emitting diode must be connected using proper current limiting diode to avoid burn outs.
- (iii) The power supply must be connected to the supply pin of the TTL chips with proper polarity and proper voltage rating.
- (iv) The TTL chips connected on the bread board must be properly fixed in order to get better connection and results.
- (v) The power supply connected must be positive five volts regulated power supply. The voltage must never exceed the tolerable limit.
- (vi) The jumper wires used to connect different pins of the integrated chips are kept short so that the base layer of the bread board doesnot get short circuited.